Engineering a Recombinant Dual-Enzyme Oral Supplement for Simultaneous Degradation of Lactose and β-Casomorphin-7 to Alleviate Dairy-Associated Gastrointestinal Symptoms

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Abstract

Although dairy intolerance is commonly attributed to lactose intolerance, growing evidence suggests that A1 β-casein may also contribute to gastrointestinal discomfort. During digestion, A1 casein releases β-casemorphin-7 (BCM-7), a bioactive opioid peptide that may produce symptoms such as delayed gastrointestinal transit, slower gut motility, and inflammatory responses through the activation of μ-opioid receptors. This paper reviews the current understanding of A1 β-casein-associated gastrointestinal intolerance, including the difference between A1 and A2 β-casein, the role of BCM-7, the challenges associated with diagnosis, and the limitations of existing management strategies. Building upon this background, a conceptual therapeutic approach is proposed that exploits the lactose-inducible lac operon during recombinant enzyme production to coordinate expression of a microbial BCM-7 degrading peptidase - DPP4 - in the presence of dairy consumption. Although this proposal remains theoretical, it presents a biologically plausible framework for developing a targeted enzymatic approach to reduce active BCM-7 during digestion. If successfully developed, this strategy could provide an alternative to dietary restriction and contribute to future therapeutic options for individuals experiencing dairy intolerance.

Introduction

Although dairy-associated discomfort is most commonly attributed to lactose intolerance, lactose intolerance represents only one of several conditions that may contribute to adverse reactions following dairy consumption. Milk protein is made up of 80% casein and 20% whey, and it generally contains two types of casein: A1 and A2. There has been an increase in the awareness of other factors that contribute to dairy intolerance, such as A1 β-casein intolerance or allergy. A1 β-casein digestion is associated with increased gastrointestinal (GI) transit time and some inflammatory markers compared with A2 β-casein (Barnett et al., 2014). Common symptoms of A1 β-casein intolerance include eczema, diarrhea, and bloating, which while similar to lactose intolerance, differ in reaction mechanisms and response times. Due to an increase in the population that experiences dairy intolerance and the limited awareness towards casein intolerance, it is necessary to develop a strategy to counter the harmful effects that casein may cause. A1 β-casein releases β-casomorphin-7 (BCM-7), which activates μ-opioid receptors in the GI tract and may contribute to the symptoms (Pal et al., 2015). Dipeptidyl peptidase 4 (DPP-4) is the primary human enzyme known to efficiently break down BCM-7 into X-Pro dipeptide at the N-terminus position, which can potentially relieve the symptoms caused by the production of BCM-7 (Giribaldi et al., 2022). Current management of dairy-associated gastrointestinal symptoms primarily relies on dietary restriction or single-enzyme supplements, leaving limited options for individuals with various kinds of dairy intolerance. This paper proposes the development of a recombinant dual-enzyme oral capsule containing β-galactosidase and a microbial BCM-7-degrading peptidase - DPP4 - as a targeted therapeutic strategy to simultaneously address multiple mechanisms underlying dairy-associated gastrointestinal discomfort.

Background

Figure 1. The evolution history of A1 and A2 milk. Adapted from (Prasad MG et al., 2024). Created with the assistance of ChatGPT.

Bovine milk is valued by many because of its nutritional value, including its contribution to 28 of the 29 nutrients that humans need (de Vasconcelos et al., 2023). Milk is composed mainly of milk proteins, which consist of casein and whey protein. Specifically, casein is further categorized into four subtypes: αs1-casein, αs2-casein, β-casein, and κ-casein (Dmour & Taha, 2018). Each of the four subtypes has dozens of different genetic variants. This paper focuses on A1-β-casein, which is one of the genetic variants of β-casein (Zinßius et al., 2024). Another genetic variant of β-casein, A2-β-casein, is considered to be the ancestral form of A1-β-casein, because it existed before the point mutation - from proline to histidine - that caused the creation of A1-β-casein in some European herds 5,000-10,000 years ago (Pal et al., 2015). *Figure 1* illustrates the evolutionary divergence of A1 and A2 β-casein variants, suggesting that A2 casein emerged before A1 casein.

The presence of histidine allows the enzyme to perform a cleavage of β-CN, which in turn causes the release of β-casomorphin-7 (BCM-7) (Giribaldi et al., 2022). Casomorphin can be defined as any opioid released from casein during digestion (Woodford, 2021). The proline-rich sequence makes it resistant to proteases, resulting in some intact BCM-7 left behind digestion, which can then bind to the μ-opioid receptors located in the central nervous system and gastrointestinal tracts of humans, delaying GI transit time and decreasing gut motility by reducing intestinal contractions (Asledottir et al., 2019). Moreover, BCM-7 is also known to increase mucus secretion in the gastrointestinal system, contributing to excessive mucus that disrupts normal intestinal function.

Ecem Bolat et al. (2024) reviewed experimental evidence indicating that BCM-7 may activate μ-opioid receptors, increase intestinal permeability, and stimulate the production of pro-inflammatory cytokines, including IL-6 and TNF-α. However, the authors also emphasized that current evidence remains insufficient to establish a direct causal relationship between BCM-7 and human diseases.

Although considerable research has investigated the relationship between A1 β-casein, BCM-7, and gastrointestinal discomfort, the results are inconclusive. Therefore, current management strategies primarily rely on dietary restriction rather than targeted interventions. This highlights the need for alternative approaches that address BCM-7 directly.

Differentiating Lactose Intolerance, Milk Allergy, and A1 β-casein Intolerance

Although the gastrointestinal symptoms associated with A1 β-casein intolerance may resemble those of lactose intolerance, these conditions differ in their underlying mechanisms. Lactose intolerance results from lactose deficiency and carbohydrate malabsorption, whereas the mechanism of A1 β-casein intolerance remains under investigation and has been hypothesized to involve bioactive peptides such as BCM-7. Distinguishing these conditions is important because they require different diagnostic approaches and management strategies.

Condition	**Cause**	**Trigger**	**Mechanism**	**Diagnostic Test**
Lactose intolerance	Lactase deficiency	Lactose	Carbohydrate malabsorption	Breath Test
Milk Allergy	Immune response	Milk proteins	IgE/non-IgE immune reaction	Allergy testing
A1 β-casein intolerance	Under investigation	Potentially BCM-7	Opioid/inflammatory response	No validated clinical test

Table 1. This table compares and contrasts the cause, trigger, mechanism, and current diagnostic test of lactose intolerance, milk allergy, and A1 β-casein intolerance.

Lactose intolerance is an enzymatic reaction that can cause symptoms such as abdominal pain, diarrhea, and bloating. People with lactose intolerance don't have enough lactase to digest lactose, leaving some lactose undigested, which is then fermented by bacteria in the large intestine. Lactose intolerance can now be detected by a breath test at most hospitals.

Conversely, a milk allergy is an adverse immune reaction to milk proteins, and most common in infants and children. Three types of reaction mechanisms can be presented in a milk allergy: IgE-mediated allergies, delayed-onset non-IgE cell-mediated allergies, and a mix of both (Hao et al., 2025). Milk proteins can act as antigens, which bind to antibodies and stimulate a series of immune responses. Symptoms include red, blotchy skin that can itch, difficulty breathing, tearing or redness of the eyes, and so on (Spergel, 2025).

Finally, A1 β-casein intolerance is a kind of dairy intolerance that is still being investigated. While the appearance of symptoms of lactose intolerance is often rapid, it can take up to 72 hours for the symptoms of casein intolerance to show. Symptoms often include bloating, stomach cramps, eczema, and rashes (Casein, n.d.).

Structural and Biochemical Differences Between A1 and A2 β-Casein

Compared to conventional milk, studies have shown that bovine milk containing only A2 casein doesn't usually generate gastrointestinal discomfort. *Figure 2* demonstrates the different intermediates that A1/A2 casein release upon enzymatic cleavage, which can be attributed to the amino acid substitution in Position 67. Since A1 casein has His instead of Pro, it's easier to be broken down by enzymes, which contributes to the release of the shorter intermediate BCM-7.

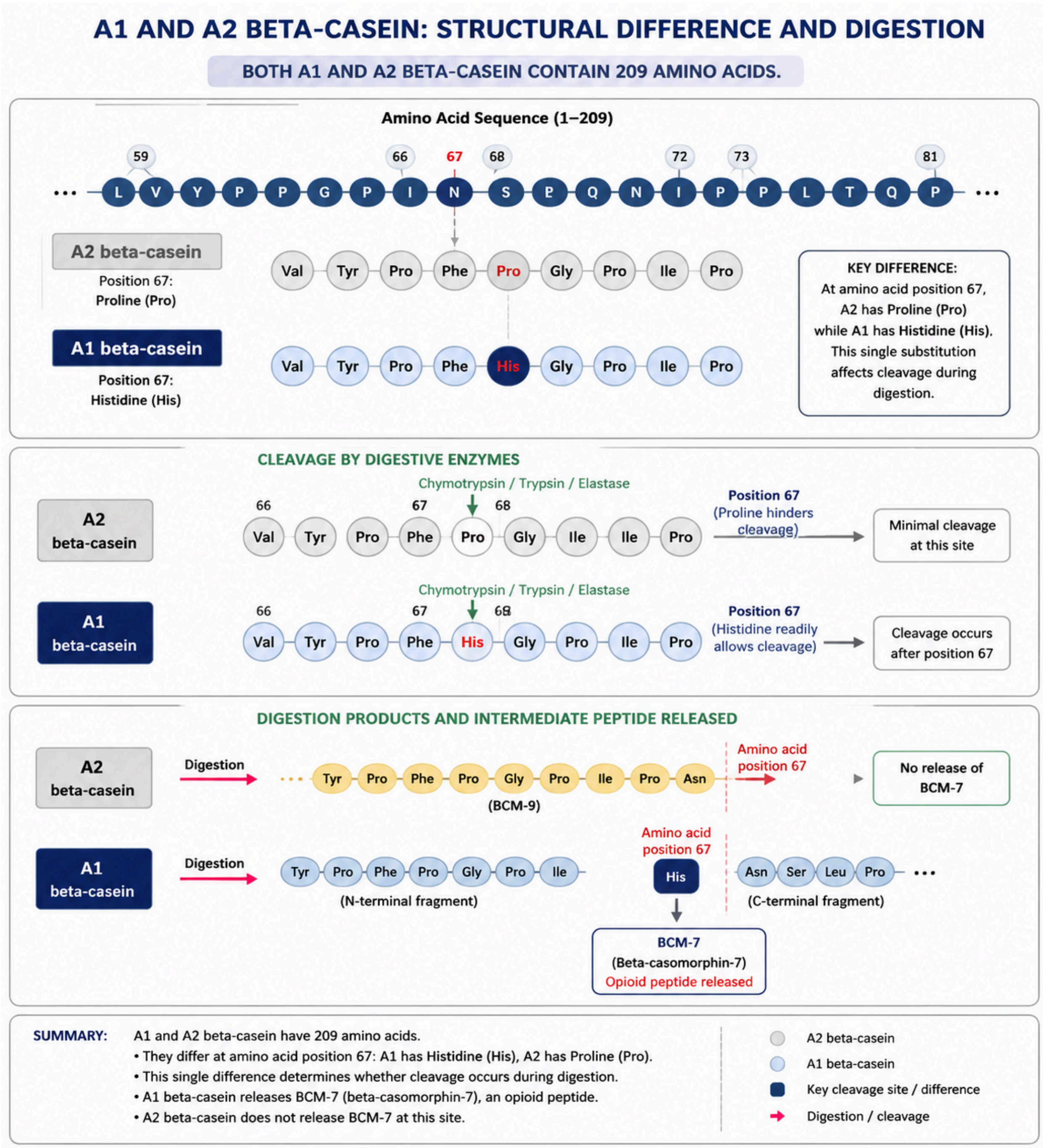

Figure 2. The structural differences between A1 and A2 casein and how it contributes to different intermediates upon enzymatic cleavage. This figure was modeled after (Pal et al., 2015) and (Prasad MG et al., 2024). Created with the assistance of ChatGPT.

While BCM-7 is released during the breakdown of A1 casein, a longer peptide, BCM-9, is released when A2 casein is digested. BCM-7 and BCM-9 are both opioid, but they perform different functions: BCM-7 is linked to inflammatory and digestive disorders such as bloating and irritable bowel patterns (University, 2024), while BCM-9 is considered beneficial, potentially both antihypertensive and antioxidant (Woodford, 2021).

Potential Mechanisms Underlying A1 Casein-Associated Intolerance

Currently, there is no confirmed cause of A1 casein intolerance. It is important to note that much research in this field is inconclusive, and no microbiological cause of A1 casein intolerance has been proven as the single explanation. However, there are some hypotheses that could be made between the enzymatic activities or gut permeability and biome and the symptoms of A1 casein intolerance. For example, the lack of DPP4 activity can contribute to impaired degradation of BCM-7. Variations in DPP4 activities may influence BCM-7 persistence in the GI tract and therefore represent a potential factor contributing to individual differences in tolerance of A1 casein. Asledottir et al., (2019) use GI fluids and porcine jejunal brush border membrane (BBM) peptidases to study the digestion of synthetic BCM-7. Results demonstrate that 5% of BCM-7 remained intact after 24 hours BBM digestion. However, the percentage of BCM-7 that persists after digestion can vary among individuals, and this variability can contribute to different responses from A1 casein, because the intact BCM-7 can interact with the opioid receptors in the gut, contributing to gastrointestinal discomfort.

Furthermore, increased gut permeability also serves as a potential contributor to inflammatory responses towards casein. A possible explanation is that since BCM-7 can trigger inflammation, it can lead to possible barrier dysfunction and thus increased gut permeability. Brooke-Taylor et al., (2017) use rodent models to demonstrate that the ingestion of A1 casein can be linked to the initiation of inflammatory responses. Inflammation and gut permeability reinforce each other, forming a cyclic pathway that attenuates the immune system. As a result, a “leaky gut” can form and contribute to increased exposure of BCM-7 to the gut opioid receptors, which might explain the gastrointestinal discomfort associated with the ingestion of A1 casein.

Current Management Strategies and Their Limitations

Current methods for casein intolerance are limited, with only some medical testing for casein allergy but not intolerance. Moreover, unlike lactose intolerance - which can be diagnosed using hydrogen breath tests; or milk allergy - which can be evaluated using skin prick tests, serum IgE tests, or oral food challenges, there is currently no validated laboratory or clinical test for A1 β-casein-associated gastrointestinal intolerance. Individuals may undergo food sensitivity testing to determine whether he or she is intolerant to dairy products, but still cannot specify the kind of intolerance. At present, suspected A1 β-casein intolerance is usually evaluated indirectly. If people still exhibit gastrointestinal discomfort after consuming dairy products and being negative in milk allergy tests and lactose intolerance tests, one can try eliminating dairy products in their diets to see whether there are other components in dairy products that may have caused the discomfort.

There are laboratory tests for milk protein allergy but none for casein intolerance, as it is more of a digestive problem rather than immune system reactions. Testing of milk allergy usually involves IgE antibodies, and it is helpful for individuals who suspect they have a milk allergy (Test, 2019). However, as previously noted, A1 β-casein intolerance cannot be revealed with this laboratory test; therefore, there is still room for improvements regarding methods of detecting milk intolerance.

The biggest challenge is that casein intolerance symptoms are largely nonspecific. Most of the symptoms in casein intolerance are also common in lactose intolerance, milk allergy, and other gastrointestinal disorders. Because of this overlap, patients are frequently misdiagnosed or assume they have lactose intolerance when the underlying cause may be different.

One approach to reduce the discomfort associated with A1 casein is to replace conventional milk with A2 milk. In contrast to traditional bovine milk, A2 milk contains only A2 casein, which cuts off the production of BCM-7. By eliminating the dietary source of BCM-7, individuals with intolerance may experience reduced symptoms related to A1 casein. For example, Jianqin et al. (2016) conducted a randomized crossover study in adults with self-reported milk intolerance and found that consumption of milk containing only A2 β-casein resulted in significantly lower gastrointestinal discomfort and reduced inflammatory markers compared with conventional milk containing both A1 and A2 casein. However, this approach does not address situations in which individuals consume conventional dairy products, such as cheese, yogurt, and cream. Consequently, while A2 milk represents a potential solution, it does not provide a therapeutic approach for individuals who are unable or unwilling to avoid conventional milk and dairy products.

Proposal

Design Rationale

Dipeptidyl peptidase 4 (DPP4) is a proline-specific serine protease that can break down proteins containing proline or alanine in the N terminus next to last (P1) position, resulting in altered function of the original protein (Aljumaah et al., 2025). Human dipeptidyl peptidase 4 (hDPP-4) plays a critical role in homeostasis - it regulates metabolic and systemic processes and modulates the functionality of over 40 bioactive substrates. BCM-7, as previously noted, is rich in proline and can be degraded by DPP4 enzymes, thus might prevent the gastrointestinal symptoms caused by the release of BCM-7 from A1 casein digestion. As a result, degrading BCM-7 serves as a potential strategy to improve health conditions while maintaining individuals' dietary flexibility.

A major challenge in developing a targeted therapy for A1 casein intolerance is identifying an appropriate biological signal. Indeed, BCM-7 that is produced during casein digestion can serve as a signal, but current research suggests that there are no known transcription factors that BCM-7 could bind to in order to initiate the production of degrading enzymes. Although it would be difficult to detect BCM-7, detecting lactose is much easier because lactose binds to the repressor that inactivates the lac operon. Given that casein's presence is often accompanied by lactose, it is reasonable to assume that if structural genes that code for hDPP4 are inserted downstream of the structural genes encoding for β-galactosidase, the presence of lactose can contribute to the production of more DPP4 enzymes. It is important to note that although DPP4 is capable of degrading BCM-7, it does not hydrolyze lactose. Therefore, β-galactosidase remains necessary to digest lactose and alleviate symptoms for those who are lactose intolerant.

Proposed Genetic Design

The lac operon was discovered in *E. coli*, meaning it naturally regulates the expression of *lacZ* - which produces β-galactosidase - *lacY*, and *lacA*, under the control of one promoter (Khan Academy, 2021). Because lactose acts as the inducer, the lac operon provides a convenient mechanism for regulating enzyme production only after dairy consumption, thereby reducing unnecessary metabolic burden on the engineered bacteria. Moreover, *E. coli* is relatively easy to experiment with, as it grows rapidly, and requires simple food (*E. Coli as a Model Organism: Advantages & Limitations | Boster Bio*, 2024). For these reasons, *E. coli* was selected as the host organism for the proposed system. The recombinant construct incorporates the coding sequence of the human DPP4 gene (*Figure 3*) downstream of the same lactose-inducible promoter, resulting in the engineered operon shown in *Figure 4*.

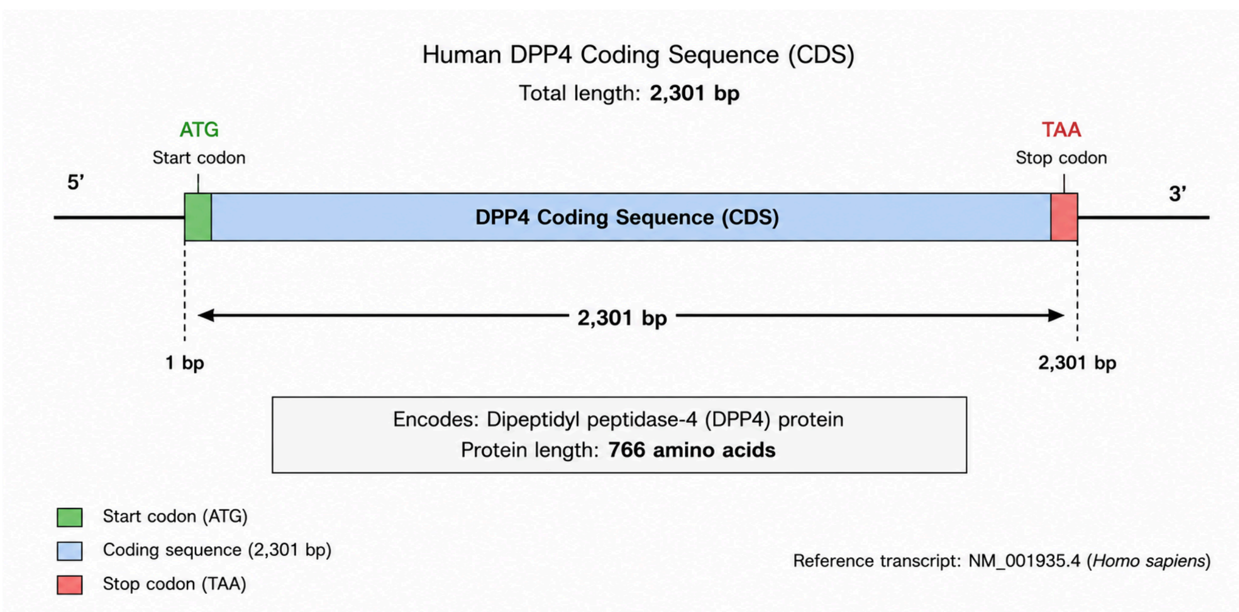

Figure 3. Coding sequence of the human DPP4 gene (CDS). Remodeled after (Dipeptidyl Peptidase 4, 2026). Created with the assistance of ChatGPT.

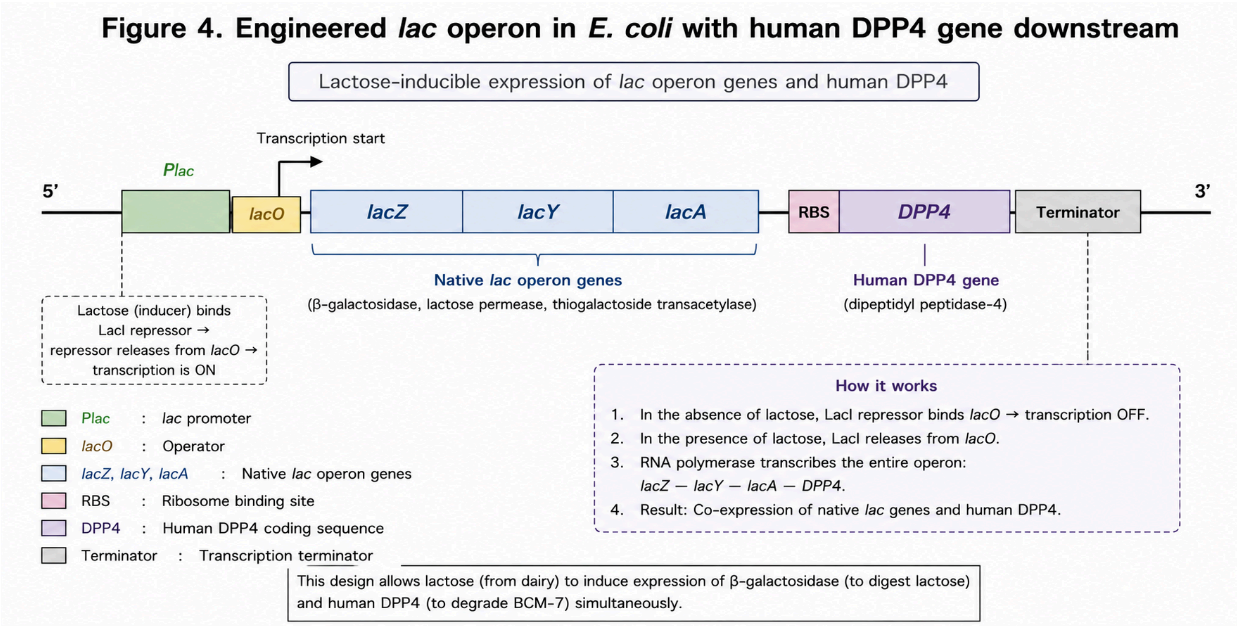

Figure 4. Engineered lac operon in E. coli with human DPP4 gene. The genetic construct is designed for E. coli. Created with the assistance of ChatGPT.

By positioning the DPP4 coding sequence downstream of the native lac structural genes, expression of DPP4 is coordinated with the natural induction of the lac operon. Consequently, whenever lactose is present following dairy consumption, both β-galactosidase and DPP4 are produced simultaneously. This coordinated regulation enables the engineered bacteria to degrade lactose while producing BCM-7-degrading enzymes under the same regulatory mechanism.

Therapeutic Applications

Following successful expression, both recombinant enzymes would be purified using standard protein purification techniques before being used. The proposed therapeutic is designed as an oral enzyme supplement containing a combination of purified β-galactosidase and DPP4 enzyme. The capsule should be consumed immediately before or together with dairy products to provide enzymatic actions during digestion and would target two potential sources of dairy-associated gastrointestinal discomfort, as individuals who experience from dairy intolerance usually acquire more than one source of intolerance.

Because BCM-7 is released into the intestinal lumen during digestion, the engineered system would require DPP4 to be secreted or released from the production strain prior to formulation (de Vasconcelos et al., 2023). Alternatively, the therapeutic product could consist of purified recombinant DPP4 produced through microbial fermentation rather than live engineered bacteria. As the proposed therapeutic contains purified recombinant enzymes rather than live genetically modified microorganisms, potential environmental release and microbial colonization risks would be substantially reduced.

To manufacture the oral capsule, a microbial production strain would be genetically engineered to co-express the genes encoding β-galactosidase and hDPP4 under the control of a lactose-inducible regulatory system.

A full workflow should look like Figure 5:

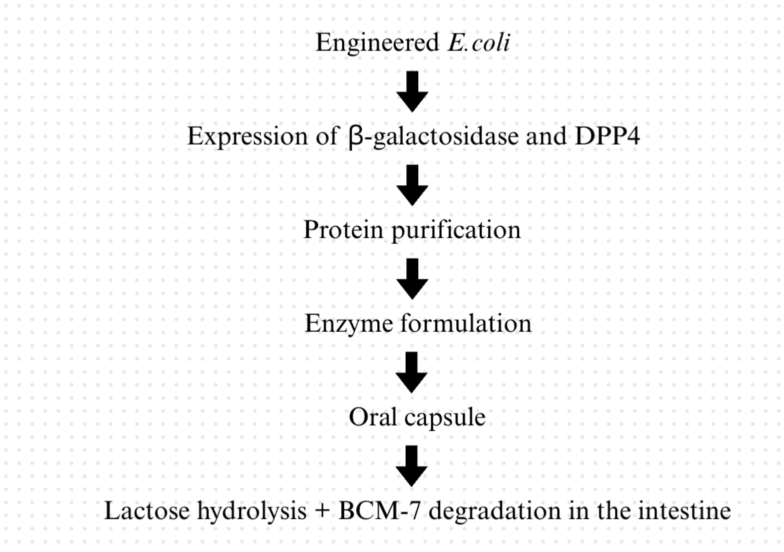

Figure 5. This figure demonstrates the designed workflow of the production and consumption of the dual-enzyme supplement. Created using canva.

Advantages

One of the major advantages of this proposed therapeutic is that it simultaneously addresses two potential factors that contribute to symptoms related to intolerance. Current lactase supplements only hydrolyze lactose, so some individuals might still experience discomfort even after taking lactase. By combining purified β-galactosidase with BCM-7-degrading peptidases, this dual-enzyme supplement has the potential to improve lactose digestion while reducing BCM-7 activities. This integrated approach may provide broader symptom relief than single-enzyme therapies.

Moreover, the production of recombinant enzymes using microbial fermentation is already widely employed in the biotechnology industry. Recombinant enzymes are already widely manufactured through microbial fermentation for pharmaceutical and food application. For example, recombinant β-galactosidase is routinely produced using microbial expression systems for commercial lactase supplements, demonstrating the feasibility of large-scale enzyme production and purification (Oliveira et al., 2011). Therefore, it presents a well-established and accessible pathway to produce designated enzymes.

Challenges and Limitations

One of the major limitations regarding this approach is that the relationship between BCM-7 and gastrointestinal discomfort remains controversial and inconclusive. While some studies suggested a potential association between BCM-7 and the symptoms, other well-controlled investigations have failed to demonstrate a consistent or significant result. Additionally, the effects of degrading BCM-7 have not yet been explored and analysis of data too has not yet been well established. The overall effectiveness of the proposed supplement therefore depends on further clarification of BCM-7's contribution to symptom development.

Another challenge is engineering an enzyme that efficiently degrades BCM-7 while minimizing activity toward physiologically important endogenous peptides. For example, studies have shown that DPP4 inhibits incretins, such as glucagon-like peptide I and glucose-dependent insulinotropic peptide, which further prevents insulin secretion that may interfere with blood sugar regulation (Bayanati et al., 2024). Future studies would therefore be required to identify or engineer enzymes with improved substrate specificity before clinical application.

Finally, it is important to recognize the potential difficulty of expressing the human DPP4 gene in *E. coli*. Although *E. coli* is widely used as a host for recombinant protein production, differences between human and bacterial gene expression systems can significantly reduce the efficiency and functionality of the expressed protein. Human genes often contain a wider variety of codons, which can lead to inefficient translation and reduced protein yield unless the coding sequence is codon-optimized. Therefore, in order to develop a more usable therapeutic, it is necessary to perform codon optimization or consider the use of alternative microbial hosts that may support a eukaryotic protein production process.

Conclusion

Since the domestication of cattle, humans have consumed milk as a major source of nutrition throughout life. Thus, detecting casein intolerance is difficult yet crucial regarding health and fitness. Many people value milk, seeing it as a beneficial choice to maintain good health and support body development; however, undesired reactions to milk can cause detrimental effects, such as chronic gastrointestinal distress and systemic inflammation, which can lead to chronic diseases.

While milk is scientifically proven to provide various benefits, the underlying costs are still being explored. For example, Chia et al., 2017 presents a potential correlation between the increased rate of Type 1 diabetes and heightened milk consumption: In Shanghai, the incidence among children aged ⩽15 years increased at a rate of 14.2% per year between 1997 and 2011, which is mirrored by an increase in per capita dairy product consumption among urban residents of 12 kg from nearly 6 kg in 1992 to 18 kg by 2006. Although both environmental factors and genetic factors play important roles in developing Type 1 diseases, this research team argues that the consumption of A1 milk is the key environmental trigger.

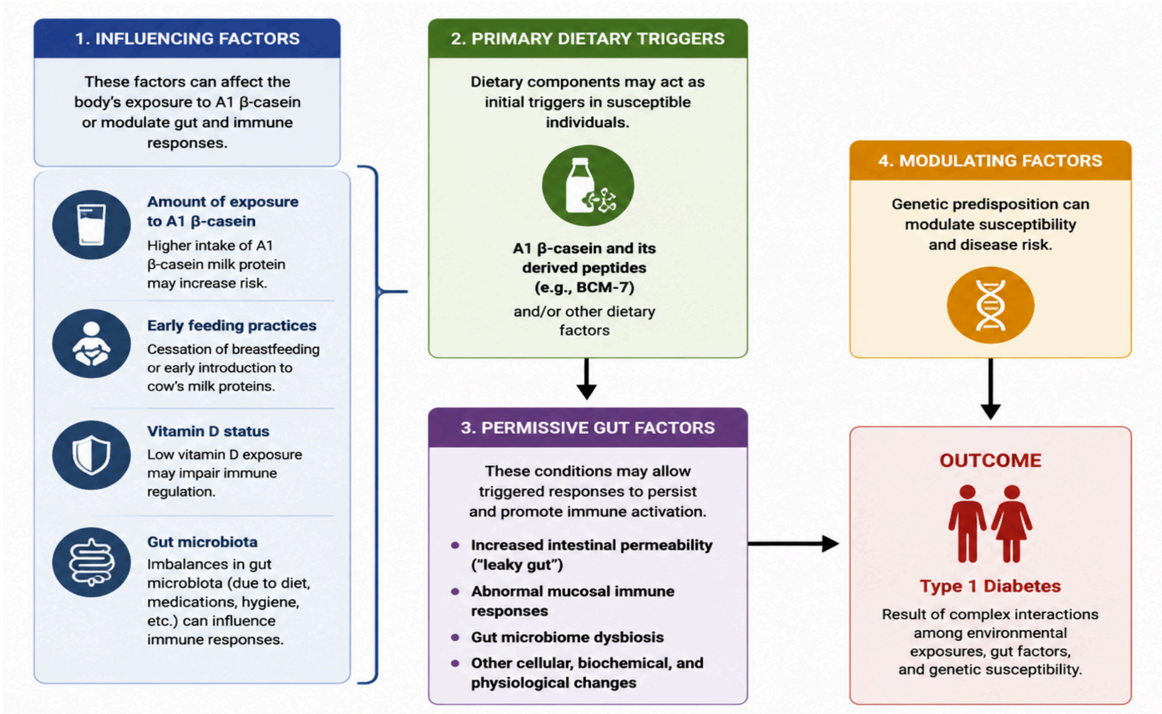

Figure 6. Proposed model for progression to Type 1 diabetes in genetically susceptible people. Adapted from (Chia et al., 2017). Created with the assistance of ChatGPT.

This proposal thus offers a novel therapeutic strategy for addressing dairy-associated gastrointestinal discomfort by simultaneously targeting lactose and BCM-7. Future experimental studies should focus on validating the catalytic efficiency of recombinant DPP4 toward BCM-7, evaluating enzyme stability under gastrointestinal conditions, and determining whether dual-enzyme supplementation provides better clinical outcomes compared with conventional lactase supplements. If successfully developed, this dual-enzyme supplement could offer individuals with greater dietary flexibility while reducing reliance on strict dairy avoidance, ultimately contributing to improved quality of life and expanding future therapeutic options for dairy-associated gastrointestinal intolerance.

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